Estimated Annual Agricultural Pesticide Use
Minidoka County, Idaho 2014-2018

24 July 2026

Summary

Minidoka County ranks **#5 of Idaho's 44 counties** by total estimated agricultural pesticide mass applied (EPest-high method, summed 2014-2018), making it one of ID's five highest pesticide-use counties. Minidoka County, in Idaho's Magic Valley near Rupert, grows irrigated potatoes, sugar beets, and beans alongside dairy operations.

Table 2 lists the rank and best available estimates of mass applied (kg) for the top 80 agricultural pesticides in **Minidoka County, Idaho** during 2018, as reported by USGS Estimated Annual Agricultural Pesticide Use data. The table also includes rank by 3-year average (2016-2018) and 5-year average (2014-2018) estimates of mass applied, and each compound's chemical class (abbreviated; see the key in Table 1 and the full classification, with a separate functional class column, in `data/pesticide_classification.csv`, shared with every chapter in this repository). Rankings and estimates are based on the *EPest-high method* described by Thelin and Stone (2013) and Baker and Stone (2015):

> "EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by-crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD." *-Pesticide National Synthesis Project webpage*

USGS lists several caveats of these data, in particular:

- Reliability decreases with scale (e.g. "detailed interpretation of where and how much use occurs within a county is not appropriate").
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates.
- A blank cell in the source file (no reported/estimated use for that compound-year-county) is treated as zero in this chapter.
- **2019 is excluded from this ranking.** The 2019 county-level file available at the time of writing covers only 69 compounds nationwide, versus roughly 400 compounds in every other year from 1992-2018 — a partial/preliminary release, not a real drop in the number of pesticides in use. Anchoring the ranking on it would silently omit hundreds of real compounds. This chapter uses 2018 (the most recent complete year) as the 1-year snapshot instead; re-run with 2019 -included windows once a complete 2019 release is available.

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)

Code	Chemical class
CPX	Chlorophenoxy compound
DIP	Dipyridyl compound
DTC	Dithiocarbamates
INO	Inorganic compounds
MIC	Microbial
CB	N-methyl carbamates (AChE inhibitor)
OGM	Organo-metallic compound
OC	Organochlorine compound
OP	Organophosphorous compound
OTH	Other

Table 1: Chemical class abbreviation key (used in the Class column of Table 2) *(continued)*

Code	Chemical class
PYR	Pyrethrins/Pyrethroids
TC	Thiocarbamates
TRZ	Triazines

Table 2: Top 80 EPest-high Estimates, Minidoka County, ID, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
1	METAM	DTC	646374	348780	328457
2	METAM POTASSIUM	DTC	416564	321713	242918
3	DICHLOROPROPENE	OC	187309	96175	115283
4	SULFURIC ACID	INO	157834	346030	311922
5	CHLOROPICRIN	OC	132844	52770	39589
6	GLYPHOSATE	OTH	65046	61688	56344
7	CHLOROTHALONIL	OTH	20706	20899	18594
8	DIMETHENAMID & DIMETHENAMID-P	OTH	15084	14166	12966
9	DIMETHENAMID-P	OTH	15084	13442	12532
10	PENDIMETHALIN	OTH	14881	13602	11777
11	MANCOZEB	DTC	14603	13292	18638
12	METRIBUZIN	TRZ	10089	9800	8967
13	EPTC	TC	8824	10930	9523
14	SULFUR	INO	7474	17499	15161
15	CHLORPYRIFOS	OP	6907	13524	11589
16	MALEIC HYDRAZIDE	OTH	6223	4085	5502
17	PYRIMETHANIL	OTH	6174	4310	3602
18	ACETOCHLOR	OTH	5827	2209	2092
19	DIQUAT	DIP	4278	3041	2446
20	PHOSPHORIC ACID	INO	3648	5107	7071
21	METHOMYL	CB	3418	2897	2286
22	METOLACHLOR & METOLACHLOR-S	OTH	3194	3048	3875
23	BROMOXYNIL	OTH	3008	2827	3716
24	METOLACHLOR-S	OTH	2551	2474	3153
25	QUINTOZENE	OTH	2466	2291	2291
26	FLUOPYRAM	OTH	2277	1505	1234
27	GLUFOSINATE	OTH	2225	987	715
28	MCPA	CPX	2198	2737	3694
29	ATRAZINE	TRZ	2177	2810	1733
30	BOSCALID	OTH	1946	3441	3005
31	FLUROXYPYR	OTH	1881	1928	1810
32	MEFENOXAM	OTH	1774	904	758
33	IMIDACLOPRID	OTH	1591	2872	3168
34	AZOXYSTROBIN	OTH	1454	1303	1304
35	PROPARGITE	OTH	1409	1409	2327
36	DICAMBA	OTH	1372	917	666
37	HEXAZINONE	TRZ	1311	958	840
38	BIFENTHRIN	PYR	1251	746	685
39	2,4-D	CPX	1183	3094	3092
40	ETHALFLURALIN	OTH	1090	1987	1784
41	ZOXAMIDE	OTH	1014	1305	1241
42	PYRACLOSTROBIN	OTH	935	1140	1280
43	DIFENOCONAZOLE	OTH	898	696	897
44	MANDIPROPAMID	OTH	880	413	451
45	SPIROTETRAMAT	OTH	872	558	364
46	TRIFLURALIN	OTH	799	762	576
47	DIURON	OTH	722	815	909

Table 2: Top 80 EPest-high Estimates, Minidoka County, ID, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) *(continued)*

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
48	METOLACHLOR	OTH	643	574	723
49	FLUAZINAM	OTH	628	575	675
50	HYDROGEN PEROXIDE	INO	564	564	564
51	CLOPYRALID	OTH	499	373	392
52	HEXYTHIAZOX	OTH	430	371	306
53	ZETA-CYPERMETHRIN	PYR	424	518	491
54	DIMETHOATE	OP	400	1549	1660
55	CARFENTRAZONE-ETHYL	OTH	399	185	141
56	METIRAM	DTC	386	386	582
57	FLUXAPYROXAD	OTH	378	497	386
58	CYMOXANIL	OTH	372	260	344
59	FAMOXADONE	OTH	372	260	344
60	OXAMYL	CB	351	342	1985
61	CYHALOTHRIN-LAMBDA	PYR	288	379	322
62	CYFLUTHRIN	PYR	286	142	106
63	PROTHIOCONAZOLE	OTH	285	129	110
64	PINOXADEN	OTH	260	292	305
65	METALAXYL	OTH	253	253	563
66	TRI-ALLATE	TC	251	215	403
67	PROPICONAZOLE	OTH	232	305	586
68	THIAMETHOXAM	OTH	221	287	214
69	TEBUCONAZOLE	OTH	203	199	239
70	COPPER HYDROXIDE	INO	187	234	209
71	ABAMECTIN	OTH	167	195	212
72	ETHOFUMESATE	OTH	145	54	91
73	PARAQUAT	DIP	138	1002	732
74	FENTIN	OGM	138	71	181
75	COPPER OXYCHLORIDE	INO	130	127	129
76	SAFLUFENACIL	OTH	124	66	66
77	PICOXYSTROBIN	OTH	119	61	52
78	2,4-DB	CPX	113	124	184
79	CLETHODIM	OTH	109	212	213
80	ALPHA CYPERMETHRIN	PYR	108	83	57

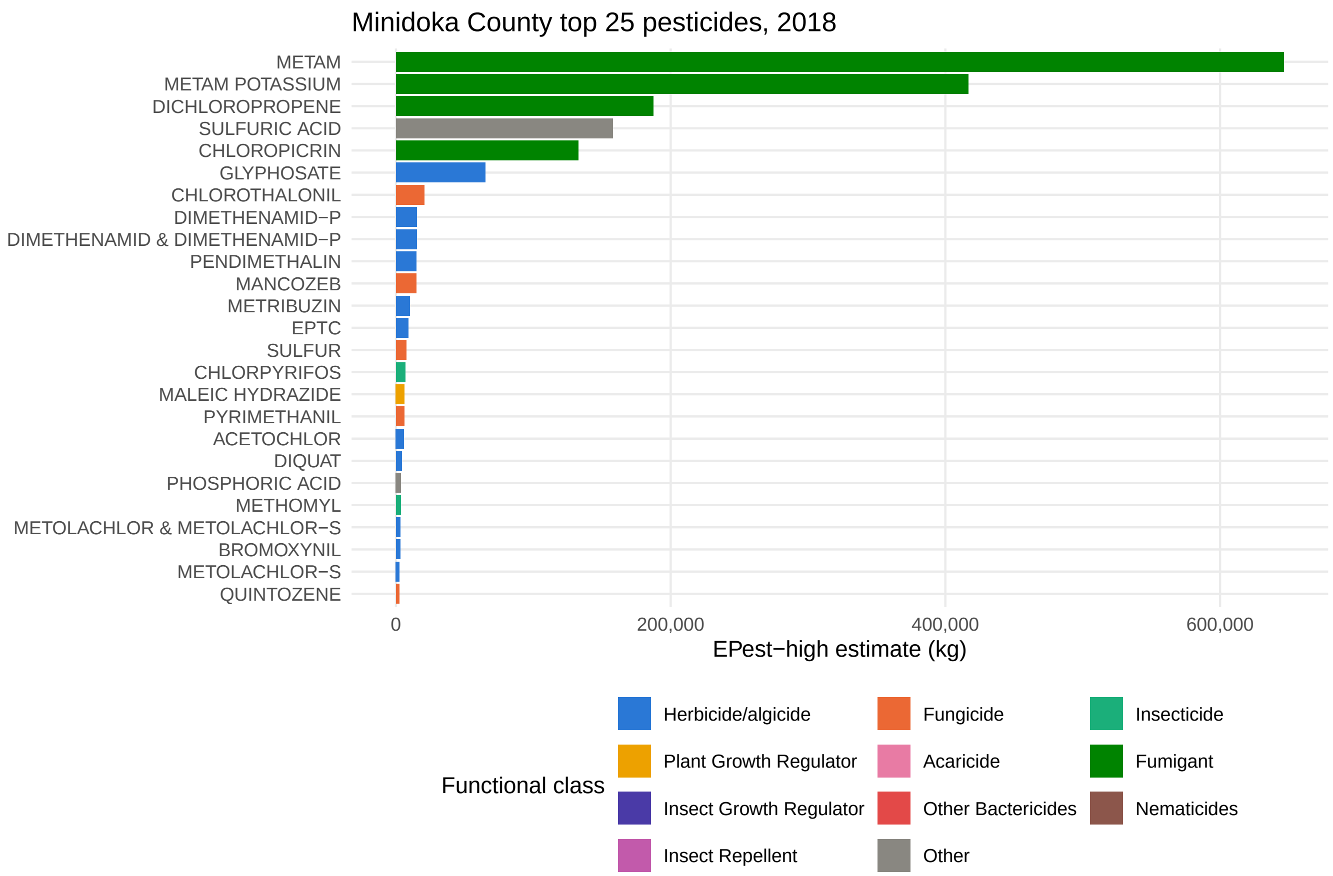

Figure 1: Top 25 compounds in Minidoka County by EPest-high mass applied, most recent year, colored by pesticide functional class.

Appendices

Table 3: Top 50 Minidoka County Estimates, Range (EPest-low and EPest-high), 1-Yr, 2018

RANK_1YR	COMPOUND	EPEST_LOW_KG_1YR	EPEST_HIGH_KG_1YR
1	METAM	646374	646374
2	METAM POTASSIUM	416564	416564
3	DICHLOROPROPENE	187309	187309
4	SULFURIC ACID	157834	157834
5	CHLOROPICRIN	1249	132844
6	GLYPHOSATE	62683	65046
7	CHLOROTHALONIL	20706	20706
8	DIMETHENAMID & DIMETHENAMID-P	12256	15084
9	DIMETHENAMID-P	12256	15084
10	PENDIMETHALIN	12933	14881
11	MANCOZEB	10133	14603
12	METRIBUZIN	9978	10089
13	EPTC	8569	8824
14	SULFUR	37	7474
15	CHLORPYRIFOS	6906	6907
16	MALEIC HYDRAZIDE	6223	6223
17	PYRIMETHANIL	6174	6174
18	ACETOCHLOR	5827	5827
19	DIQUAT	4278	4278
20	PHOSPHORIC ACID	0	3648
21	METHOMYL	24	3418
22	METOLACHLOR & METOLACHLOR-S	1928	3194
23	BROMOXYNIL	3008	3008
24	METOLACHLOR-S	1928	2551
25	QUINTOZENE	2466	2466
26	FLUOPYRAM	2277	2277
27	GLUFOSINATE	2225	2225
28	MCPA	2198	2198
29	ATRAZINE	1	2177
30	BOSCALID	1946	1946
31	FLUROXYPYR	1855	1881
32	MEFENOXAM	1774	1774
33	IMIDACLOPRID	1591	1591
34	AZOXYSTROBIN	1220	1454
35	PROPARGITE	0	1409
36	DICAMBA	537	1372
37	HEXAZINONE	0	1311
38	BIFENTHRIN	1251	1251
39	2,4-D	920	1183
40	ETHALFLURALIN	1090	1090
41	ZOXAMIDE	1014	1014
42	PYRACLOSTROBIN	935	935
43	DIFENOCONAZOLE	898	898
44	MANDIPROPAMID	880	880
45	SPIROTETRAMAT	872	872
46	TRIFLURALIN	26	799
47	DIURON	0	722
48	METOLACHLOR	0	643
49	FLUAZINAM	628	628
50	HYDROGEN PEROXIDE	0	564

Table 4: Top 50 Minidoka County Estimates, Range (EPest-low and EPest-high), 3-Yr Avg, 2016-2018

RANK_3YR	COMPOUND	EPEST_LOW_KG_3YR_AVG	EPEST_HIGH_KG_3YR_AVG
1	METAM	348780	348780
2	SULFURIC ACID	346030	346030
3	METAM POTASSIUM	321713	321713
4	DICHLOROPROPENE	96175	96175
5	GLYPHOSATE	59878	61688
6	CHLOROPICRIN	1033	52770
7	ETHOPROPHOS	0	22316
8	CHLOROTHALONIL	20718	20899
9	PROPAMOCARB HCL	0	19040
10	SULFUR	15	17499
11	DIMETHENAMID & DIMETHENAMID-P	12499	14166
12	PENDIMETHALIN	11821	13602
13	CHLORPYRIFOS	13465	13524
14	DIMETHENAMID-P	12499	13442
15	MANCOZEB	11699	13292
16	EPTC	10845	10930
17	METRIBUZIN	9650	9800
18	PHOSPHORIC ACID	3891	5107
19	PYRIMETHANIL	4310	4310
20	MALEIC HYDRAZIDE	4085	4085
21	BOSCALID	3441	3441
22	2,4-D	2664	3094
23	METOLACHLOR & METOLACHLOR-S	2020	3048
24	DIQUAT	3041	3041
25	METHOMYL	18	2897
26	IMIDACLOPRID	2872	2872
27	BROMOXYNIL	2827	2827
28	ATRAZINE	95	2810
29	MCPA	2701	2737
30	METOLACHLOR-S	2020	2474
31	PETROLEUM OIL	2440	2440
32	QUINTOZENE	2291	2291
33	ACETOCHLOR	2209	2209
34	DIMETHENAMID	0	2172
35	ETHALFLURALIN	1411	1987
36	FLUROXYPYR	1769	1928
37	FLUTOLANIL	1669	1669
38	DIMETHOATE	471	1549
39	FLUOPYRAM	1505	1505
40	PROPARGITE	0	1409
41	ZOXAMIDE	1305	1305
42	AZOXYSTROBIN	1190	1303
43	PYRACLOSTROBIN	932	1140
44	NALED	0	1059
45	PARAQUAT	39	1002
46	GLUFOSINATE	987	987
47	HEXAZINONE	0	958
48	DICAMBA	416	917
49	MEFENOXAM	904	904
50	DIURON	0	815

Table 5: Top 50 Minidoka County Estimates, Range (EPest-low and EPest-high), 5-Yr Avg, 2014-2018

RANK_5YR	COMPOUND	EPEST_LOW_KG_5YR_AVG	EPEST_HIGH_KG_5YR_AVG
1	METAM	268302	328457
2	SULFURIC ACID	311922	311922
3	METAM POTASSIUM	201514	242918
4	DICHLOROPROPENE	57721	115283
5	GLYPHOSATE	54565	56344
6	CHLOROPICRIN	786	39589
7	PROPAMOCARB HCL	0	19040
8	MANCOZEB	17683	18638
9	CHLOROTHALONIL	18486	18594
10	ETHOPROPHOS	0	15192
11	SULFUR	539	15161
12	DIMETHENAMID & DIMETHENAMID-P	10924	12966
13	DIMETHENAMID-P	10924	12532
14	PENDIMETHALIN	10540	11777
15	CHLORPYRIFOS	11500	11589
16	EPTC	9153	9523
17	METRIBUZIN	8801	8967
18	PHOSPHORIC ACID	2335	7071
19	MALEIC HYDRAZIDE	5502	5502
20	METOLACHLOR & METOLACHLOR-S	2454	3875
21	BROMOXYNIL	3304	3716
22	MCPA	3666	3694
23	PYRIMETHANIL	3602	3602
24	IMIDACLOPRID	3168	3168
25	METOLACHLOR-S	2377	3153
26	2,4-D	2736	3092
27	BOSCALID	3005	3005
28	DIQUAT	2446	2446
29	PROPARGITE	0	2327
30	QUINTOZENE	2291	2291
31	METHOMYL	16	2286
32	DIMETHENAMID	0	2172
33	ACETOCHLOR	2092	2092
34	OXAMYL	1789	1985
35	TERBUFOS	1886	1886
36	FLUROXYPYR	1674	1810
37	ETHALFLURALIN	1439	1784
38	ATRAZINE	105	1733
39	DIMETHOATE	597	1660
40	FLUTOLANIL	1518	1518
41	PETROLEUM OIL	1471	1473
42	AZOXYSTROBIN	1217	1304
43	PYRACLOSTROBIN	918	1280
44	ZOXAMIDE	1241	1241
45	FLUOPYRAM	1234	1234
46	NALED	0	1059
47	DIURON	0	909
48	DIFENOCONAZOLE	677	897
49	HEXAZINONE	0	840
50	SODIUM CHLORATE	0	780